

Recurrent Deep Learning for EEG-based Motor Imagination Recognition

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Abstract—Deep Learning has grasped great attention for recognition of Electroencephalography. For the analysis of brain dynamics, non-stationary motor imagery signals are used. Although a number of studies have been carried out for the extraction of hidden patterns and classification of EEG signals, temporal information has rarely been incorporated. In this paper, we propose a spatio-temporal energy maps generation scheme followed by deep learning classification model. Common spatial pattern filters and Fast Fourier Transform Energy Maps are deployed to obtain discriminative and spatio-temporal features. Long-Short-Term-Memory (LSTM) based neural network has been proposed to classify the temporal series of energy maps. This research also investigates preprocessing techniques to obtain optimal parameters which include frequency bands selection and temporal segmentation. The proposed model is evaluated on BCI Competition IV dataset 2a and achieved 0.64 mean kappa for multi-class EEG classification, which is the current state of the art. Furthermore, several empirical findings are also presented, that may be of significant interest to the BCI community.

Keywords—Artificial Intelligence, Deep Learning, Convolutional Neural Network, Electroencephalography

I. INTRODUCTION

BCI is an emerging technology which connects the tangible engine cortex range of human cerebrum and deals with brain aided control of devices and computer. Handicapped humans, who have lost a marginal grip over motor cortex due to severe injuries can be connected with such a BCI system to restore their abilities. Brain activities related to motor imagery (MI) tasks can be obtained by the 10 – 20 cathode position framework connected in invasive or non-invasive manners. Electroencephalogram is considered as the most effective way to collect non-invasive MI tasks. BCI system empowers the patients to utilize motor imagery signals, by mapping their sensory-motor function, to control prosthetic devices, 2D cursor control [1], wheelchair control [2] and quad-copter con-

trol [3]. Therefore, great efforts have been made to effectively classify motor imagery signals [4]. In the last decade, various publications have been accounted for the classification of MI signals using Convolutional Neural Network (CNN). In [5], the authors utilized Steady State Visually Evoked Potential (SSVEP) in which visual stimulus was utilized with CNN of four layers followed by a Fourier Transform layer. They contended that conversion of temporal data into frequency domain results in better classification accuracy.

Deep Belief Networks (DBNs) are another form of CNN which learn the patterns as per layer such that current layer works as the input for the next layer [6] [7] [8] [9]. A few pieces of literature have also focused on DBNs to classify motor imagery signals. To classify emotions, in [10], a DBN was employed and fed with the differential entropy features computed on EEG signals. To capture a reliable emotional stage switching, a Hidden Markov Model was utilized. Another deep learning method was reported to classify two class motor imagery tasks [11]. A DBN classifier was trained on a single channel data and later on combined multiple restricted Boltzmann Machines (RBMs) as a stack using AdaBoost algorithm. To train the DBN, they used Contrastive Divergence (CD) algorithm and reported that they achieved better performance than SVM classifiers. Similarly, a new deep learning scheme based on RBM was proposed in [12]. To obtain frequency domain features, they used Wavelet Packet Decomposition (WPD) and fast Fourier transform. Four layer classifier was proposed by stacking three RBMs at the top of each other and a *Softmax* regression as the final prediction layer and the network was named Frequential Deep Belief Network (FDBN). Conjugate gradient and back-propagation were used to fine tune the network. FDBN was evaluated on benchmark datasets and significant performance improvement has been shown in comparison to the state-of-the-art methods.

However, the EEG signals contains very small in amplitude, which arises a major challenge to extract the features and classify the multiclass EEG data. Hence artifacts like muscular movement, eye movement, eye blink, teeth grinding variations

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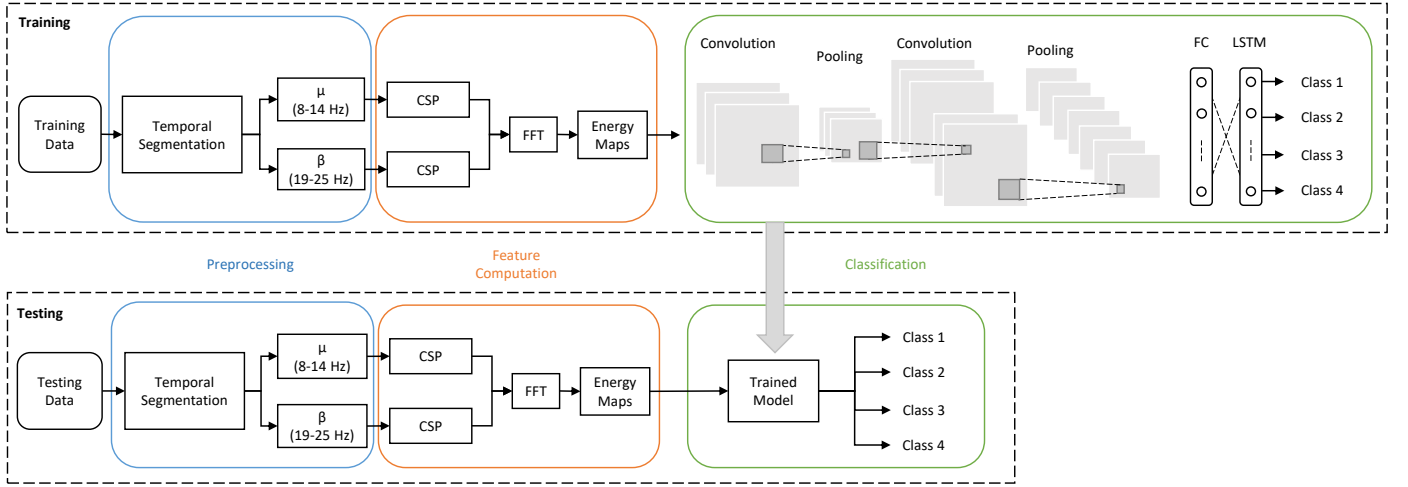


Fig. 1. Proposed Model, Input: 4 class motor imagery data, output: class confidence.

in heart rhythm affects the signal to noise ratio (SNR) and weak values of SNR can introduce classification challenges. Further, utilization of time and frequency based features are reported in the literature but feeding these feature as a temporal stream has not reported yet. Training a model in such a way that it must remember the previous temporal input to learn a sequential pattern can increase the classification performance. To address this issue, we propose a novel approach that retains the information of previous samples. In contrast to all the previous deep learning based methods, we incorporate temporal information of previous samples rather than training the classifier only on the current sample.

II. METHODOLOGY

Proposed deep learning model consists of three sequential stages: (i) data preprocessing, (ii) feature extraction and, (iii) classification. Structure of the proposed model is shown in Figure 1. In the preprocessing stage, temporal trimming, temporal segmentation, frequency bands selection and normalization are performed. To yield the best results, considerable attention has been paid to all preprocessing techniques and near-optimal parameters are selected from an exhaustive list of experimental variants. To find most discriminative features and to ensure maximum variance among all the features, CSP filters are used. These discriminative features are transformed into energy maps using Fast Fourier Transform Energy Maps (FFTEM) and used as the final feature set for the classifier. LSTM based Convolutional Neural Network is proposed as a classifier to retain the temporal information and classify a sequence of temporal segments.

A. Preprocessing

Reduction of artifacts is important in the preprocessing stage as artifacts in raw EEG data can affect classification performance. Temporal trimming of the occurrence of actual motor imagery task and corresponding frequency ranges are also important factors. A time based window is utilized to trim each EEG trial such that trimmed samples contain most effective patterns related to motor imagery. To form a temporal stream, temporal segmentation is applied by dividing each

trial into a stream of chunks. Hence, each chunk may contain different ranges of amplitudes so each chunk of EEG signal is normalized to balance the data ranges.

1) *Temporal Trimming and Segmentation*: In practice, an entire window of each trial is used for feature extraction. Previous experiments have shown that the effectiveness of feature extraction and classification performance is highly dependent on subject-specific time segments of EEG signals [13]. In order to address this issue, overlapping time segments of different time duration ranges between 3.0-6.5 are tried out to extract near-optimal range of each trial. To make the temporal sequence of a single trial, we further trimmed each trial, obtained from temporal trimming stage, into multiple chunks where each chunk contains 0.1 sec interval of the trimmed trial. By dividing a trial in this way, we obtained 20 temporal segments as a sequence.

2) *Frequency Bands Selection*: The occurrence of motor imagery task can vary in a frequency range between 8Hz to 30Hz and highly depends on the subjects. As per prior knowledge, two frequency bands, μ (8-13 Hz) and β (13-30 Hz) are filtered as these bands are reported as the most effective band ranges in literature [14]. For filtration of these bands, a Butterworth filter of 5th order is used and resulted in two frequency bands (μ , β).

3) *Normalization*: After trimming, segmenting and filtering the data, normalization is applied on each segment to remove the constant bias. For each session and each electrode i , the mean $\mu(x_i)$ of the signal is subtracted from every time measurement sample $x_i(t)$. The result is divided by the standard deviation $\sigma(x_i)$.

B. Feature Extraction

Filtered streams obtained from preprocessing stage are fed to feature extraction stage where cascaded features are computed. In first stage, CSP filters are applied to find discriminative spectral features such that it maximizes variance of a class and at the same time minimizes the variance of all other classes. Variance maximization and minimization is achieved by determining spatial patterns via simultaneous

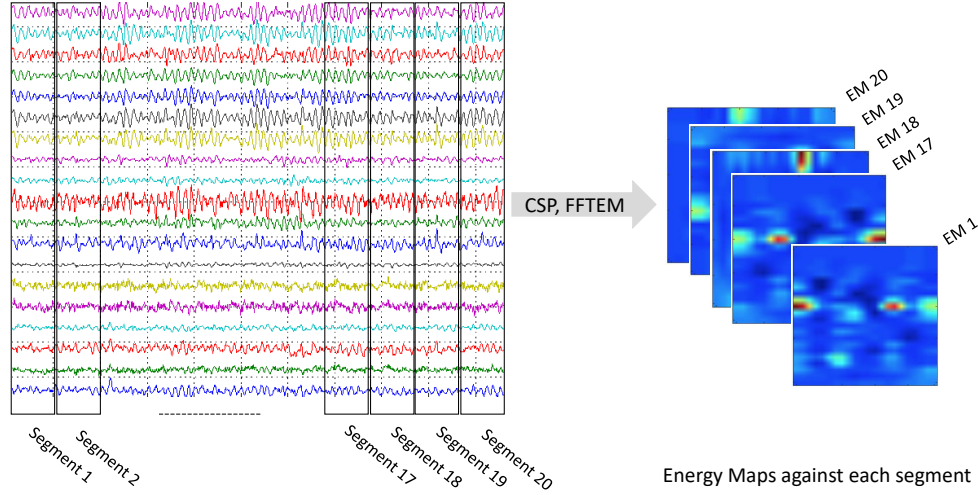


Fig. 2. Temporal segmentation and sequence of FFT Energy Maps (EM).

diagonalization of covariance matrices for all classes. These discriminative features are used to generate FFT energy maps to capture hidden spectral and temporal information.

1) *Energy Maps*: In addition to spectral features, energy density computation on spectral features can result in more prominent features of an event. To extract the spectral information, we used the Fast Fourier Transform, from which the energy maps are obtained and resulted as Fourier Transform Energy Maps. For the generation of 2D (energy vs frequency) feature maps, FFTEM is a strong candidate as compared to Fourier Transform (FT), and Wavelets in terms of computational complexity, without compromising the performance [15].

Discriminative features, obtained from CSP filters, are fed to the FFTEM module to compute energy vs frequency maps against μ , β bands. Resultant maps contain information of 22 channels against μ and β frequency bands. An example of an energy map (for the entire frequency range) is shown in Fig. 2 where it can be seen that high energies are between 8 – 27 Hz and are distributed among all channels. All EEG channels are mapped on the spectral features and can be formulated as:

$$EM_i = |\text{FFT}(y_i)|, i = \overline{1, N} \quad (1)$$

where EM is the obtained 2D energy map and N is total number of segments. Y is a single CSP feature obtained from the CSP filter and as used as input to the FFTEM module. These energy maps are used as the final feature set for the classifier in both training and testing phase.

C. Classification

To incorporate temporal information of EEG signal, we introduce LSTM based CNN classifier, where the LSTM layer is introduced before the CNN classifier. For a fair comparison, we also experimented a stand-alone CNN network without LSTM layer and conventional classifiers: Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM).

To classify multi-class motor imagery EEG signals, baseline CNN is used which contains two convolutional layers,

a pooling layer after each convolution layer, fully connected (FC) layer and final classification layer (*Softmax*) as shown in Figure 1.

Convolution layer consists of non-linear filters and performs 3D convolution on the provided normalized feature maps, obtained from FFTEM module. Filters of a convolutional layer learn discriminative features hidden in energy maps and learn the patterns and update weights of the model on each iteration. Features obtained from convolutional layer may contain a large number of positive or negative values which can lead some neurons to learn the patterns effectively. To solve this issue, an activation function is used to map the convolution features on a predefined interval. There is not a single activation function which suits best to a specific problem so we evaluated multiple activation functions to find the optimal activation function for the given problem and found *ReLU* as the most optimal activation function.

Convolution layer results in a large number of features which can increase the computational complexity. To reduce the feature size, pooling layer is used. Pooling layer window of size 2×2 with a stride of 2 in both dimensions is applied over all convolutional features and pooling layer resulted in maximum value from each window and discarded the remaining three entries. The final *Softmax* layer computes a probability to map the input on all classes and the maximum probability corresponding to the predicted class label.

Recurrent neural networks are reported as a good choice to retain information of long term events. However, RNN can face “gradient vanish or explode” problem during back propagation with an increase in dependencies on the previous samples [16]. EEG signals, measured by neural activity, can be regarded as dynamic non-linear temporal series and recurrent networks like LSTM networks appear to be an effective method of choice to retain the temporal information of EEG signals and can reduce the “gradient vanish or explode” problem. A temporal stream of energy maps of discriminative features, obtained from FFTEM module, is fed to the LSTM network to learn the patterns against each class. LSTM layer contains three gate structures, i.e., input gate, memory gate and forget

gate. These gates determine which chunk of prior information should be added into the main data flow and which information should be discarded. At each gate, an activation function is applied to map the data onto a nonlinear function (*Sigmoid*). These functions are denoted as input activation (f_i), forget gate activation (f_g) and output activation (f_o) and generate decision vectors. For example, input gate generates a decision vector i_t with given prior hidden state h_{t-1} and a current input x_t . Generation of i_t can be formalized as;

$$\mathbf{i}_t = f_g(w_i h_{t-1} + w_i x_t + b_i) \quad (2)$$

where b_i is the bias w_i is the weighted input matrix. X_t and H_{t-1} can be used to generate input candidate information as in equation:

$$\mathbf{C}_t = f_i(w_c h_{t-1} + w_c x_t + b_c) \quad (3)$$

Forget gate (f_g) generates decision vector f_t which determines the factor of the prior unit state C_{t-1} to be reserved. f_t can be formulated as:

$$\mathbf{f}_t = f_g(w_h h_{t-1} + w_x x_t + b_f) \quad (4)$$

Vector f_t scaled between 0 and 1 using *Sigmoid* function. Output state of LSTM cell is obtained from output gate f_o and can be formulated as:

$$\mathbf{o}_t = f_g(w_o h_{t-1} + w_o x_t + b_o) \quad (5)$$

To obtain the best number of LSTM layers and units in each layer, we experimented with up to three layers and 32 to 256 numbers of memory cells and evaluated on unseen EEG data (which are summarized in section III). The model is estimated using the fast and efficient Adam optimizer, a mini-batch gradient descent variant with categorical cross-entropy (CE) as the loss function. The learning rate for Adam was set to 0.001 and decay rates of first and second moments of 0.9 and 0.999, respectively. We used 10-fold cross-validation strategy for evaluation and measure mean kappa value (mean κ).

To make a fair comparison, we also tried some other classifiers on the obtained features (FFTEMs) including Linear Discriminant Analysis (LDA), Support Vector Machine (SVM) and CNN. LDA and SVM classifiers are binary classifiers and to use them for a multiclass problem, we applied One-vs-Rest (OVR) technique and selected the class which classifies the test data with the greatest margin without changing other parameters.

D. Dataset Configuration

In this paper, we used BCI Competition dataset 2a [14] which consist of four class motor imagery signals including left hand, right hand, both feet, and tongue data. The dataset was collected in a controlled environment from 9 subjects on two different days and 288 trials were conducted on each day using 22 EEG channels and 3 EOG channels. The recording was based on a cue where a fixation cross was shown with a beep to imagine one of the four MI actions (left hand, right hand, both feet, and tongue). There was a short break between the recording of each trial. Trials with artifacts were discarded to make dataset less noisy and additional correction of artifacts reduction were performed by the dataset developers. Training and evaluation sessions were recorded separately. Cohen's Kappa (k) is recommended for evaluation by BCI competition

TABLE I. COMPARISON OF CLASSIFIERS IN TERMS OF κ COEFFICIENTS ON DATASET 2A BCI COMPETITION IV.

Subjects	Classifiers						
	LDA	SVM	CNN	LSTM			
				No. of Memory Cells			
				32	64	128	256
S1	0.61	0.64	0.70	0.68	0.70	0.71	0.68
S2	0.45	0.47	0.45	0.42	0.42	0.47	0.45
S3	0.67	0.56	0.76	0.63	0.69	0.84	0.81
S4	0.63	0.71	0.58	0.59	0.62	0.64	0.60
S5	0.77	0.72	0.69	0.48	0.51	0.56	0.52
S6	0.45	0.48	0.54	0.62	0.71	0.68	0.65
S7	0.49	0.55	0.63	0.53	0.55	0.59	0.61
S8	0.51	0.56	0.56	0.60	0.62	0.64	0.59
S9	0.55	0.53	0.59	0.61	0.58	0.63	0.66
Mean κ	0.57	0.58	0.61	0.57	0.60	0.64	0.62

IV committee, hence is used for evaluation of the proposed model.

III. RESULTS AND DISCUSSION

In this section, we present our experimental results on different classifiers and compare our results with the techniques in literature.

A. Comparison of Classifiers

In this research, we used LDA, SVM and CNN classifiers along with the proposed LSTM classifier for multi-class motor imagery classification. A comparison of all classifiers is summarized in Table III. LDA classifier is used in OVA mode whereas kernel and soft-margin hyperparameters are tuned to get best results from the SVM classifiers. It can be seen that LDA and SVM classifiers performed very close to each other but SVM surpasses the LDA performance in terms of mean κ . CNN classifier result in 0.61 mean κ with a standard deviation of 0.095. In contrast to CNN classifier, the final proposed model (LSTM) resulted in 0.64 mean κ with a standard deviation of value 0.109.

B. Influence of Memory Cells

In terms of memory cells, large numbers can increase the model complexity while a small number can concise the memory environment. To find the optimal number of memory cells and LSTM layers, a range of memory cells has been evaluated. Obtained results are summarized in Table III. Model performed best with a single layer and 128 LSTM memory cells, which is consistent with results obtained in [24] (classified ERP signals). More layers and a larger number of LSTM memory cells improved training accuracy but led to strong overfitting and a performance decrease on the validation sets. Less than 128 memory cells led to underfitting, which was observed by a decrease in both training and validation accuracy.

C. Comparison with other state-of-the-art Methods

Performance of the proposed method is compared with state of the art methods and summarized in Table II. Kappa values against each subject are computed and mean κ values are computed against each method. In [23], researchers

TABLE II. COMPARISON OF CLASSIFICATION ACCURACY (KAPPA COEFFICIENTS) ON DATASET 2A BCI COMPETITION IV.

Subjects	LSTM	NNMF [17]	CNN [18]	LTCSP [19]	FBCSP [20]	TI [21]	PW [22]	OVR [22]	DC [22]	SCSSP [23]
S1	0.71	0.70	0.70	0.74	0.68	0.88	0.78	0.68	0.71	0.62
S2	0.47	0.45	0.45	0.43	0.43	0.22	0.40	0.42	0.37	0.28
S3	0.84	0.71	0.76	0.80	0.70	0.88	0.75	0.75	0.66	0.66
S4	0.64	0.96	0.58	0.57	0.61	0.39	0.52	0.48	0.41	0.33
S5	0.56	0.60	0.71	0.31	0.80	0.53	0.41	0.40	0.40	0.14
S6	0.68	0.43	0.51	0.38	0.42	0.33	0.19	0.27	0.26	0.25
S7	0.59	0.55	0.63	0.76	0.55	0.38	0.80	0.77	0.73	0.41
S8	0.64	0.61	0.56	0.75	0.55	0.85	0.74	0.75	0.58	0.60
S9	0.63	0.57	0.60	0.80	0.56	0.81	0.54	0.61	0.50	0.66
mean κ	0.64	0.62	0.61	0.61	0.59	0.58	0.57	0.57	0.52	0.44
standard deviation	0.11	0.19	0.10	0.20	0.12	0.27	0.21	0.18	0.17	0.19

proposed separable common spatio-temporal patterns (SCSSP) for multi-class motor imagery classification and achieved 0.44 mean κ . In [21], researchers were able to achieve 0.58 mean κ by using KNN and SVM classifiers for information fusion and transductive learning. In [23], researchers introduced Separable Common Spatio-Spectral Pattern (SCSSP) for four class EEG data and reported 0.44 mean κ performance. The achieved classification accuracy was not promising as compared to other techniques. Divide and Conquer (DC), Parsing Window (PW) and One versus Rest (OVR) classification methods were tried out in where the OVR and PW methods resulted the best performance by achieving 0.57 mean κ [22].

Local Temporal Common Spatial Pattern (LTCSP) for feature extraction and Linear Discriminant Analysis (LDA) for classification was proposed in [19] and resulted 0.61 mean κ . However, 0.62 mean κ was reported in [17] where researchers used Non-Negative Matrix Factorization (NNMF) and SVM for feature extraction and classification of multiclass EEG signals, respectively. However, the proposed methodology utilize CSP for discriminative feature extraction, FFTEM for spectro-temporal feature mapping and energy maps generations LSTM based network for sequential classification. Furthermore, it is found to be more robust as compared to [17][19][20] methods by comparing variance between the subjects.

IV. CONCLUSION

A novel sequence to sequence learning method using LSTM is proposed for multiclass motor imagery classification. In order to extract discriminative features, CSP and FFTEM algorithms are applied to generate spectro-temporal energy maps. A Long-Short-Term-Memory based CNN model is proposed to classify four class motor imagery signals. To learn sequential sequence, temporal segmentation is applied to break single trial into sequential chunks. In contrast to the reported methods, the proposed methodology can extract the spatial, spectral and temporal hidden features encoded in EEG data to learn the complex patterns. FFTEM features are computed against each trial for each subject after extracting discriminative features using CSP to maximize co-variance among all four classes. Extensive experimental evaluation of each module has been carried out on a publicly available benchmark dataset and yielded the state of the art results. Experimental results demonstrated that the proposed method might be useful for high-performance motor imagery classification applications.

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REFERENCES

- [1] Y. Li, J. Long, T. Yu, Z. Yu, C. Wang, H. Zhang, and C. Guan, "An eeg-based bci system for 2-d cursor control by combining mu/beta rhythm and p300 potential," *IEEE Transactions on Biomedical Engineering*, vol. 57, no. 10, pp. 2495–2505, 2010.
- [2] J. Long, Y. Li, H. Wang, T. Yu, J. Pan, and F. Li, "A hybrid brain computer interface to control the direction and speed of a simulated or real wheelchair," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 20, no. 5, pp. 720–729, 2012.
- [3] K. LaFleur, K. Cassady, A. Doud, K. Shades, E. Rogin, and B. He, "Quadcopter control in three-dimensional space using a noninvasive motor imagery-based brain-computer interface," *Journal of neural engineering*, vol. 10, no. 4, p. 046003, 2013.
- [4] F. Lotte, M. Congedo, A. Lécuyer, F. Lamarche, and B. Arnaldi, "A review of classification algorithms for eeg-based brain-computer interfaces," *Journal of neural engineering*, vol. 4, no. 2, p. R1, 2007.
- [5] H. Cecotti and A. Graeser, "Convolutional neural network with embedded fourier transform for eeg classification," in *2008 19th International Conference on Pattern Recognition*. IEEE, 2008, pp. 1–4.
- [6] W. Abbas and M. Tap, "Adaptively weighted multi-task learning using inverse validation loss," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 1408–1412.
- [7] W. Abbas, M. H. Shakeel, N. Khurshid, and M. Taj, "Patch-based generative adversarial network towards retinal vessel segmentation," in *International Conference on Neural Information Processing*. Springer, 2019, pp. 49–56.
- [8] S. Ali, W. Abbas, N.-U. Hassan, A. Raza, and W. Zhang, "Cpgan: Conditional patch-based generative adversarial network for retinal vessel segmentation," *IET Image Processing*, 2019.
- [9] W. Abbas and N. A. Khan, "A high performance approach for classification of motor imagery eeg," in *2018 IEEE Biomedical Circuits and Systems Conference (BioCAS)*. IEEE, 2018, pp. 1–4.
- [10] W.-L. Zheng, J.-Y. Zhu, Y. Peng, and B.-L. Lu, "Eeg-based emotion classification using deep belief networks," in *2014 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2014, pp. 1–6.
- [11] X. An, D. Kuang, X. Guo, Y. Zhao, and L. He, "A deep learning method for classification of eeg data based on motor imagery," in *International Conference on Intelligent Computing*. Springer, 2014, pp. 203–210.
- [12] N. Lu, T. Li, X. Ren, and H. Miao, "A deep learning scheme for motor imagery classification based on restricted boltzmann machines," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 25, no. 6, pp. 566–576, 2016.
- [13] H. Azami, H. Hassanpour, J. Escudero, and S. Sanei, "An intelligent approach for variable size segmentation of non-stationary signals," *Journal of advanced research*, vol. 6, no. 5, pp. 687–698, 2015.

- [14] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. Mueller-Putz *et al.*, “Review of the bci competition iv,” *Frontiers in neuroscience*, vol. 6, p. 55, 2012.
- [15] P. Heckbert, “Fourier transforms and the fast fourier transform (fft) algorithm,” *Computer Graphics*, vol. 2, pp. 15–463, 1995.
- [16] Y. Bengio, P. Simard, P. Frasconi *et al.*, “Learning long-term dependencies with gradient descent is difficult,” *IEEE transactions on neural networks*, vol. 5, no. 2, pp. 157–166, 1994.
- [17] W. Abbas and N. A. Khan, “A discriminative spectral-temporal feature set for motor imagery classification,” in *2017 IEEE International Workshop on Signal Processing Systems (SiPS)*. IEEE, 2017, pp. 1–6.
- [18] W. Abbas and N. A. Khan, “Deepmi: Deep learning for multiclass motor imagery classification,” in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2018, pp. 219–222.
- [19] H. Ghaheri and A. Ahmadyfard, “Extracting common spatial patterns from eeg time segments for classifying motor imagery classes in a brain computer interface (bci),” *Scientia Iranica. Transaction D, Computer Science & Engineering, Electrical*, vol. 20, no. 6, p. 2061, 2013.
- [20] W. Abbas and N. A. Khan, “Fbcsp-based multi-class motor imagery classification using bp and tdp features,” in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2018, pp. 215–218.
- [21] H. Raza, H. Cecotti, and G. Prasad, “A combination of transductive and inductive learning for handling non-stationarities in motor imagery classification,” in *2016 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2016, pp. 763–770.
- [22] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter bank common spatial pattern algorithm on bci competition iv datasets 2a and 2b,” *Frontiers in neuroscience*, vol. 6, p. 39, 2012.
- [23] A. S. Aghaei, M. S. Mahanta, and K. N. Plataniotis, “Separable common spatio-spectral patterns for motor imagery bci systems,” *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 1, pp. 15–29, 2015.
- [24] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, “Learning representations from eeg with deep recurrent-convolutional neural networks,” *arXiv preprint arXiv:1511.06448*, 2015.